

SMART IRRIGATION SYSTEM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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EXTERNAL EXAMINER

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ABSTRACT

The main objective of this project is to propose a Smart IOT based Agriculture assisting farmers in getting Live Data (Temperature, Soil Moisture) for efficient environment monitoring which will enable them to do smart farming and increase their overall yield and quality of products. The Agriculture being proposed via this project is integrated with Arduino Technology, Breadboard mixed with various sensors and live data feed can be obtained online from Blynk Application.

In order to achieve this, an IoT (Internet of Thing) Module is interfaced to the Arduino Node MCU board at the receiver end while on the transmitter end, a Blynk Application on the mobile phones sends ON/OFF commands to the receiver where loads are connected. By touching the specified location on the Blynk App, the loads can be turned ON/OFF remotely through this technology. The loads are operated by IoT board through relay module.

Along with this we use soil sensor, which detects whether soil is dry or wet. When the soil condition is dry soil sensor give command to IoT module to start the pump and when soil becomes wet it gives command to stop the pump. It works in accordance with the soil condition. This project is complete smart project for advance irrigation.

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LIST OF ABBREVIATION

IOT **INTERNET OF THINGS**

GND **GROUND**

NC **NORMALLY CLOSED**

NO **NORMALLY OPENED**

COM **COMMON**

PIR **PASSIVE INFRARED SENSOR**

CHAPTER 1

INTRODUCTION

1.1 GENERAL

Agriculture plays major role in the economy of the country. More than 70% of Indian population relies on agriculture for their sustenance. As the contribution of agriculture to Gross Domestic product is declining nowadays, we are in urge to increase crop productivity with efficient and effective water usage. In agriculture irrigation is the important factor as the monsoon rainfalls are unpredictable and uncertain. Agriculture in the face of water scarcity has been a big challenge. There have been many systems developed using advanced technologies that help to prevent excessive and scarce watering to crops and thereby increase the crop yield.

There are some parameters to determine irrigation of crops. Soil moisture and temperature of the field are the most essential parameters. The electromagnetic sensors are used to detect soil moisture. This method saves 53% of water compared to sprinkler irrigation. The sensor nodes continuously sense the crop field and send it to the coordinator node. These are some methods that have been used so far to improve irrigation system [1], decrease crop wastage and increase crop productivity. In this work the system is developed using sensors to monitor crop - field. The water usage is 90% more efficient than any other traditional and other modern irrigation methods.

Smart irrigation technologies help people precisely schedule when lawns and crops need to be watered and how much water these plants require. Using IoT/M2M sensors, growers are able to monitor the soil moisture levels around their plants, as well as monitor weather conditions, so they can use water more efficiently and effectively.

The hardware and software requirements of this project include Arduino, soil moisture sensor, Wi-Fi module ESP8266, Arduino CC (IDE), Android studio, and MySQL.

1.2 FACTS ABOUT INDIAN AGRICULTURE

- About 70-72% of the total population of the country is dependent on agriculture.
- Approximately 58.2% of our population derives its livelihood from agriculture.
- It provides food to the second biggest population and the biggest population of cattle in the world.
- Our agro-based industries are fully dependent on raw material provided by agriculture.
- Agriculture with its allied activities accounts for 45% of our national income.

1.2.1 TYPE OF AGRICULTURE IN INDIA

1.2.1.1 Subsistence Farming

- The figure 1.1 shows the subsistence type of Agriculture, farmers work hard to grow enough food to survive only.
- In this type of farming the produce is consumed mainly by farmer and his family.
- There remains no surplus to sell in the market.



Figure 1.1 Subsistence Farming

1.2.1.2 Mixed Farming

- The combination of agriculture and pastoral farming is called Mixed Farming.
- In this type of farming, cultivation of crops and rearing of animals are done together on the same farm.

1.2.1.3 Jhum /Shifting Cultivation

- This is a primitive form of agriculture in which a plot of land is cultivated a few years and then is deserted.
- This slash and burn method of farming is carried on in jungles of north-eastern part of India e.g. in Assam, Meghalaya, Mizoram etc.
- A plot of land is cleared for cultivation as shown in the figure 1.2. As the yield decreases after two or three years, the plot is abandoned and a fresh clearing is made.



Figure 1.2 Shifting Cultivation.

1.2.1.4 Extensive Farming

- This is a system of farming in which the cultivator uses a limited amount of labour and capital on relatively large area.
- This type of agriculture is shown in the figure 1.3. which is practiced in countries where population's size is small, and land is enough.

- Here, per acre yield is low but overall production is in surplus due to less population.
- Agriculture is done with the help of machines.



Figure 1.3 Extensive Farming.

1.2.1.5 Intensive Farming

- The figure 1.4 shown below is a system of farming in which the cultivator uses large amount of labour and capital on a relatively small area.
- In countries where the size of population is big but land is less, this type of farming is done.
- Annually two or three crops are grown due to the demand of food for the large size of population
- Agriculture is done with the help of manual labour.



Figure 1.4 Intensive Farming.

1.2.1.6 Plantation Agriculture

- The figure 1.5 shown the type of agriculture, trees or bushes are planted on huge estates.

- A single crop like rubber, sugarcane, coffee, tea or banana is grown.
- These crops are major items of export.



Figure 1.5 Plantation Agriculture

1.2.2 PROBLEMS OF INDIAN AGRICULTURE

- The low productivity of our agriculture is mainly due to the difficulties [2] faced by our peasants.
- Indian agriculture is chiefly of subsistence type where a large manual labour is employed to work on farms to grow just enough food for the needs of the family and very little is left for marketing.
- A major part of the Indian soil has been impoverished because it has been under plough for the last 4000 or 5000 years.
- Deforestation, overgrazing and heavy rainfall have led to soil erosion.
- Divisions of land have led to fragmentation.
- The size of land holding is very small and uneconomic.
- The farmers are poor, illiterate and ignorant.
- They use primitive tool and out dated method.
- They lack financial credit and investment.
- Good seeds, fertilizers and improved technology are not available to them.
- They lack irrigation facilities and are still on the mercy of nature.

- Most of the farmers have no security against crop failure or less caused by nature.
- Generally farmers are uneducated and have no scientific approaches.
- Reporting weather conditions.

1.3 NEED FOR AGRICULTURE

- Agriculture plays a vital role in the growth of any state.
- The primary sector of an economy comprises agricultural and other activities and contributes a significant amount to the Gross Domestic Product (GDP).
- Agriculture provides raw materials for many industries which form the backbone of the nation.
- For decades, agriculture has been associated with the production of essential food crops. At present, agriculture above and beyond farming includes forestry, dairy, fruit cultivation, poultry, mushroom, etc. Today, processing, marketing, and distribution of crops and livestock products etc. are all acknowledged as part of current agriculture. Thus, agriculture could be referred to as the production, processing, promotion and distribution agricultural products. Agriculture plays a critical role in the entire life of a given economy.
- The major agricultural products can be broadly grouped into foods, fibres, fuels and raw materials (such as rubber). Food classes includes cereals (grains), vegetables, fruits, oils, meat, milk, fungi and eggs.
- Over one-third of the world's workers are employed in agriculture, second only to the service sector, although the number of agricultural workers in developed countries has decreased significantly over the centuries.

- Crop produces hardware and software Systems that measure moisture, temperature, and electrical conductivity in the soil.
- Their system tells farmers when and how much to irrigate. The soil layer of the earth is critical in maintaining plant life, offering mechanical support, and supplying water and nutrients. Soil functions as a major storage mechanism of heat, collecting energy throughout the day and releasing heat to the surface during the night. Over the course of a year, soil retains energy during warmer seasons and releases heat to the air throughout colder season.

1.4 IMPORTANCE OF MOISTURE AND TEMPERATURE PARAMETERS IN AGRICULTURE

Crop produces hardware and software Systems that measure moisture ,temperature , and electrical conductivity in the soil. Their system tells farmers when and how much to irrigate. The soil layer of the earth is critical in maintaining plant life, offering mechanical support, and supplying water and nutrients. Soil functions as a major storage mechanism of heat, collecting energy throughout the day and releasing heat to the surface during the night. Over the course of a year ,soil retains energy during warmer seasons and releases heat to the air throughout colder seasons.

- Soil temperature directly affects plant growth. In other words, nearly every crop slows down its growth when soil temperatures are below 90C and above 50 0C.
- Germination of various seeds requires different soil temperature ranges. Forexample, maize starts to germinate at soil temperatures from 7 to 100C.
- Most organisms within soil thrive at temperatures between 25-350°C.
- Nitrification requires a soil temperature of 320C.

1.5 IMPORTANCE OF SENSORS

Smart agriculture [3], also known as precision agriculture, allows farmers to maximize yields using minimal resources such as water, fertilizer, and seeds. By deploying sensors and mapping fields, farmers can begin to understand their crops at a micro scale, conserve resources, and reduce impacts on the environment. Smart agriculture has roots going back to the 1980s when Global Positioning System (GPS) capability became accessible for civilian use. Once farmers were able to accurately map their crop fields, they could monitor and apply fertilizer and weed treatments only to areas that required it. During the 1990s, early precision agriculture users adopted crop yield monitoring to generate fertilizer and pH correction recommendations. As more variables could be measured and entered a crop model, more accurate recommendations for fertilizer application, watering, and even peak yield harvesting, could be made.

1.5.1 SOIL MOISTURE SENSORS

70 percent of the world's freshwater withdrawals go towards irrigation uses. Close to 90 percent of freshwater is used for irrigation in India. It is recorded that only 5 percent of water gets used for domestic purpose in India. Despite that, farmers often obtain sub-optimal yields from their crop due to improper irrigation management practices. What if farmers could monitor soil moisture in their fields? It would help them take has informed decisions on irrigation and be cost-effective too. By monitoring soil moisture, farmers can now optimize their water usage, increase produce yield, produce high quality crops, reduce water resource degradation and save a lot of money.

Monitoring soil moisture will help farmers understand the actual soil water condition and how much their crops need to use. Soil moisture sensors enable in taking informed irrigation decisions on when to irrigate and how much water must be supplied to avoid low quality production.

With the help of soil moisture sensors, some of these questions that are commonly asked by farmers are easily addressed.

- When is the best time for irrigation?
- How do I know if enough water has infiltrated the soil?
- What is the soil depth reached by irrigation water?
- Am I using too little water? Too much or, just right?
- How much water loss can I expect due to soil evaporation?
- Has the crop roots absorbed enough water?
- During days of no irrigation, how much reserve will be available for my crop?

1.5.2 TEMPERATURE SENSOR

The temperature of farmland microclimate is taken as the research object, the temperature time series data is organized, different time granularities are divided, and the trend characteristics are analysed. The traditional time series analysis method and the neural network prediction method are used to predict the different time granularity, respectively.

Temperature series are time series data, and the commonly used methods of analysing time series data were divided into traditional time series prediction model and data-driven time series forecasting model.

1.6 IMPLEMENTATION OF SMART IRRIGATION SYSTEM

The agricultural sector is set to face enormous challenges in order to feed the 9.6 billion people that the Food and Agriculture Organization [3] (FAO) predicts are going to inhabit the planet by 2050. So what's the solution? Welcome to smart farming. Smart Farming is a farming management concept using modern technology to increase the quantity and quality of agricultural products. The farming and agricultural industry relies on innovative ideas and technological advancements to help increase yields and better allocate resources.

The late 19th century and the 20th century brought several mechanical innovations, like tractors and harvesters. Today, a driving force behind increased agricultural production at a lower cost is the Internet of Things (IOT). Internet of Things applications in agriculture include farm vehicle tracking, livestock monitoring, storage monitoring, and much more.

The next several years will see increasing use of these smart farming technologies. In fact, IOT device installations in the agriculture world are projected to experience a compound annual growth rate of 20 percent. According to a January 2016 MACHINA Research report, the number of connected agricultural devices is expected to grow from 13 million at the end of 2014 to 225 million by 2024.

1.7 SENSOR INTERFACING

Some sensors like moisture sensors [4], Temperature & Humidity Sensor, Ultrasonic Water level sensor and RTC module are embedded and require microcontrollers to interface. Creating the sensor and weatherizing it is an engineering challenge that would need to be met.

Positioning the Soil sensor for an optimal communication path is another engineering challenge. And to fixing the water level sensor in the tank if required is a challenge. The Temperature sensor can be mounted on the motor body to observe its temperature lively to analyze the condition of motor. If sensors are placed in an orange orchard, the trees may interfere if the antennae aren't mounted properly. This is obviously not as big of a challenge if an antenna is mounted in a strawberry field.

1.8 COMMUNICATION OF DATA

- The distance that data needs to travel makes a huge impact on what type of technology should be used.

- For short distances, we may use radio frequency identification (RFID) or near field communication (NFC), which is common in cell phones.
- If there is a need to send data over hundreds or even thousands of meters, lowpower, wide-area network (LPWAN) might be a better option. An application over this type of network might be used to measure soil moisture in fields or to find and track livestock as they graze.
- In conclusion, the technology we engineer into our IOT agriculture application should not be a hindrance to what the end user is trying to measure. So, make sure that we understand what the end user needs to measure then choose a solid technology to build around to get that information out.

1.9 OBJECTIVES

- The primary objective of the project is:
- To improve the crop yield by introducing IOT technology in agriculture.
- The main aim is to decrease the crop wastage and increase the crop productivity with efficient and effective water usage

1.10 SUMMARY

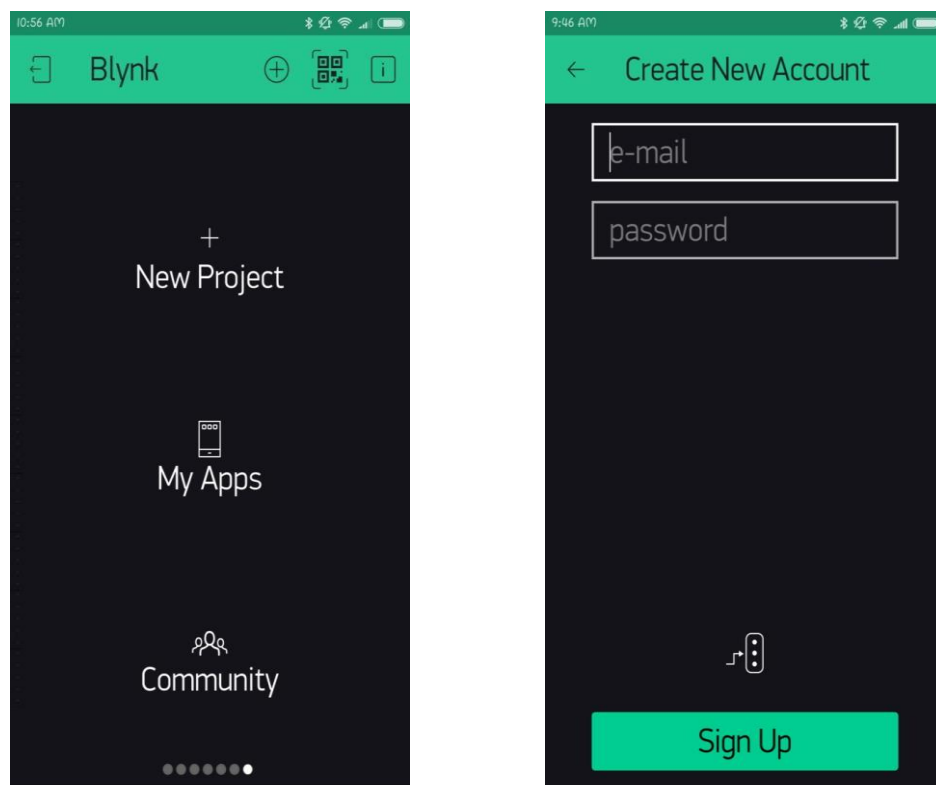
The chapter deals about the acts about the Indian agriculture system and the importance of agriculture for the growth of nation. It is also dealing about the importance of sensors for monitoring the important parameters such as temperature and moisture. The different types of sensors used for monitoring the parameter is also discussed in this chapter. The chapter also deals with the data communication system and its various technology for adopting this environment in Internet of Things (IOT). Finally, the chapter summarized with the primary objective of this project to improve the crop yield by introducing IOT technology in agriculture.

CHAPTER 2

BLYNK MOBILE APPLICATION

2.1 ABOUT BLYNK APPLICATION

Blynk is a Platform [5] with IOS and Android apps to control Arduino, Raspberry Pi and the likes over the Internet. It's a digital dashboard where you can build a graphic interface for your project by simply dragging and dropping widgets.



(a) New Project

(b) Sign Up

Figure 2.1 Creating New Project in Blynk App.

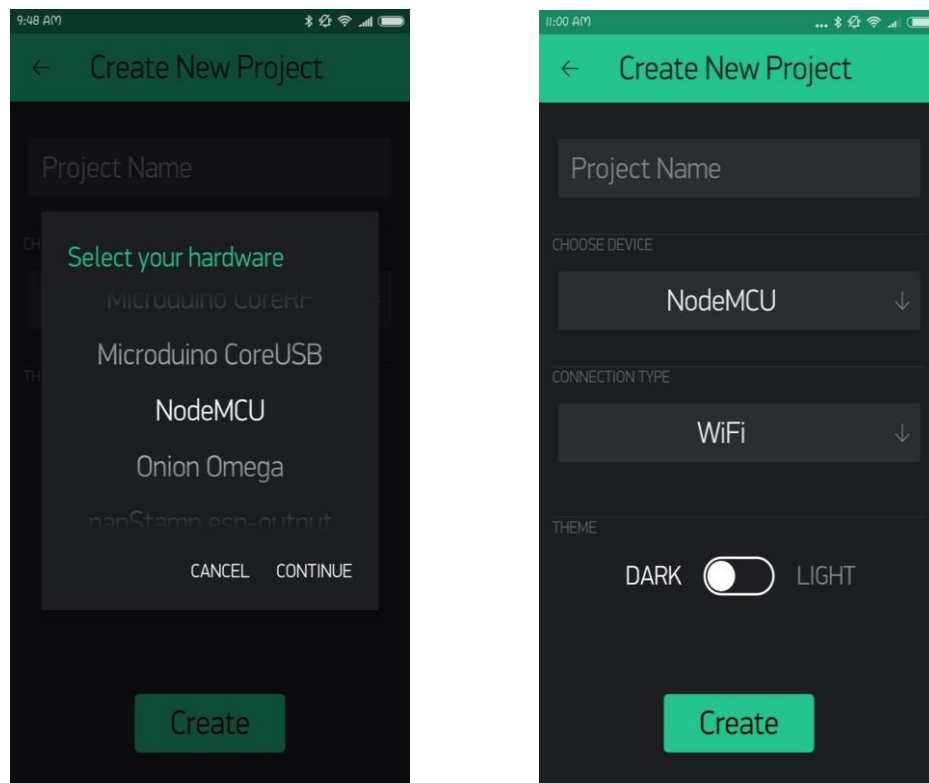
The Figure 2.1 (a) & (b) shows that how to create a new project by click a new project and communicate with each other to build a fully functional IoT application that can be controlled from anywhere through a preconfigured connectivity type. You can control your hardware from the Blynk app running on your mobile device through the Blynk Cloud or Blynk's personal server.

It works the same in the opposite direction by sending rows of processed data from hardware to your Blynk app.

2.2 CREATING A BLYNK PROJECT

Click the “Create New Project” in the app to create a new Blynk app. Give it any name.

The figure 2.2 (a) & (b) shows that how Blynk works with hundreds of hardware models and connection types. Select the Hardware type. After this, select connection type. In this project we have select Wi-Fi connectivity.



(a) Select Module

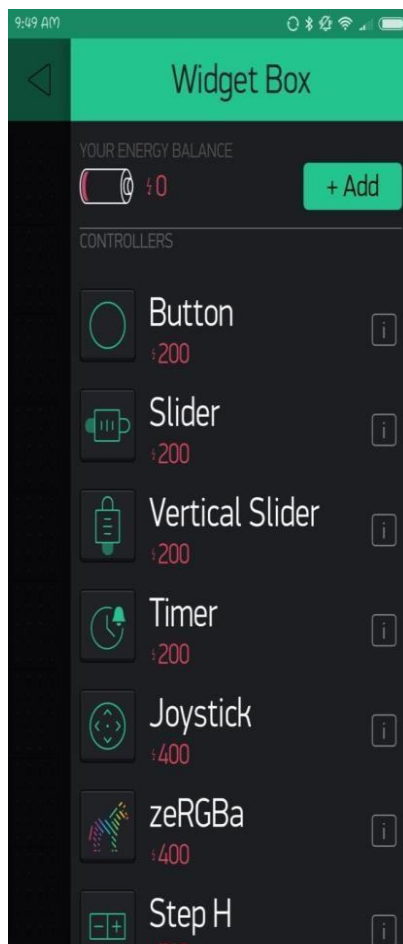
(b) Select Theme

Figure 2.2 Creating Arduino Module in Blynk App.

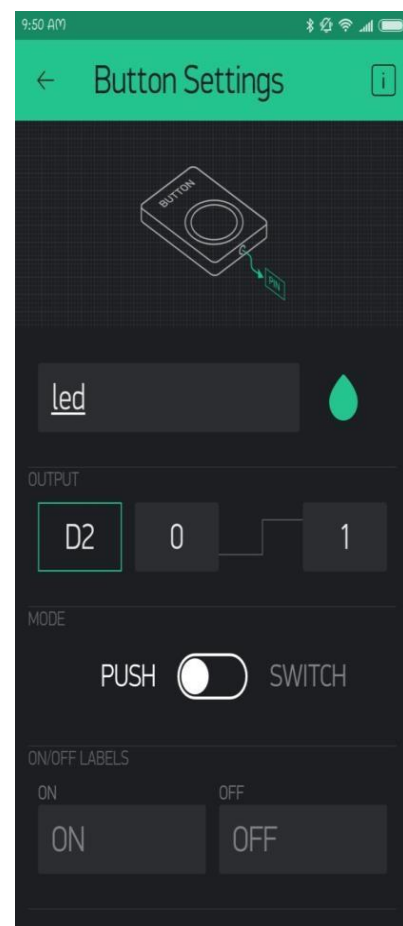
You'll also need to install the Blynk Arduino Library, which helps generate the firmware running on your ESP8266. Download the latest release from <https://github.com/blynkkk/blynk-library/releases> , and follow along with the directions there to install the required libraries.

2.3 WIDGET AND BUTTON SETTINGS

Then you'll be presented with a blank new project. The figure 2.3 (a)& (b) shown below is to open the widget box, click in the project window to open. We are selecting a button to control Led connected with NodeMCU. Click on Button. Give name to Button say led. Under OUTPUT tab- Click pin and select the pin to which led is connected to NodeMCU, here it is digital pin 2, hence select digital and under pin D2. And Click continue. Under MODE tab- Select whether you want this button as "push button" or "Switch". You have successfully created a GUI for Arduino.



(a) Select widget



(b) Button Settings

Figure 2.3 Widget and button settings in Blynk App.

2.4 STAND ALONE PROGRAMMING CODE

Before uploading, make sure to paste your authorization token into the auth [] variable. Also make sure to load your Wi-Fi network settings into the Blynk.begin (auth, “ssid”, “pass”) function.

```
#define BLYNK_PRINT

Serial #include

<ESP8266WiFi.h> #include

<BlynkSimpleEsp8266.h>

// You should get Auth Token in the

Blynk App.char auth[] =

"YourAuthToken";

// Your Wi-Fi credentials. // Set password to "" for

open networks.char ssid[] = "YourNetworkName";

char pass[] =

"YourPassword";void

setup()

{ // Debug console

Serial.begin(9600);

Blynk.begin(auth, ssid,

pass);          }

void loop()

{ Blynk.run(); }
```


CHAPTER 3

COMPONENTS AND SENSORS USED IN SMART IRRIGATION SYSTEM

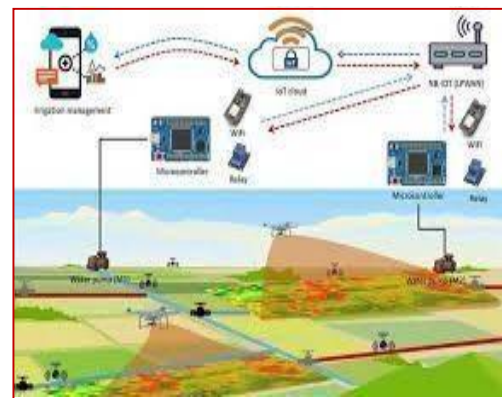
3.1. INTERNET OF THINGS

Internet of Things [6] (IoT) describes an emerging trend where many embedded devices (things) are connected to the Internet. These connected devices communicate with people and other things and often provide sensor data to cloud storage and cloud computing resources where the data is processed and analyzed to gain important insights. Cheap cloud computing power and increased device connectivity is enabling this trend.

The Internet of Things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, or people that are provided with identifiers and the ability to transfer data over a network without requiring human- to-human (or) human-to-computer interaction is shown in figure 3.1 (a).”



(a) IoT System



(b) Smart Irrigation

ExampleFigure 3.1 Internet of Things for Irrigation system.

IoT solutions are built for many vertical applications such as environmental monitoring and control, health monitoring, vehicle fleet monitoring, industrial monitoring and control and home automation is ..

3.2 Nodemcu ESP8266 Pinout:

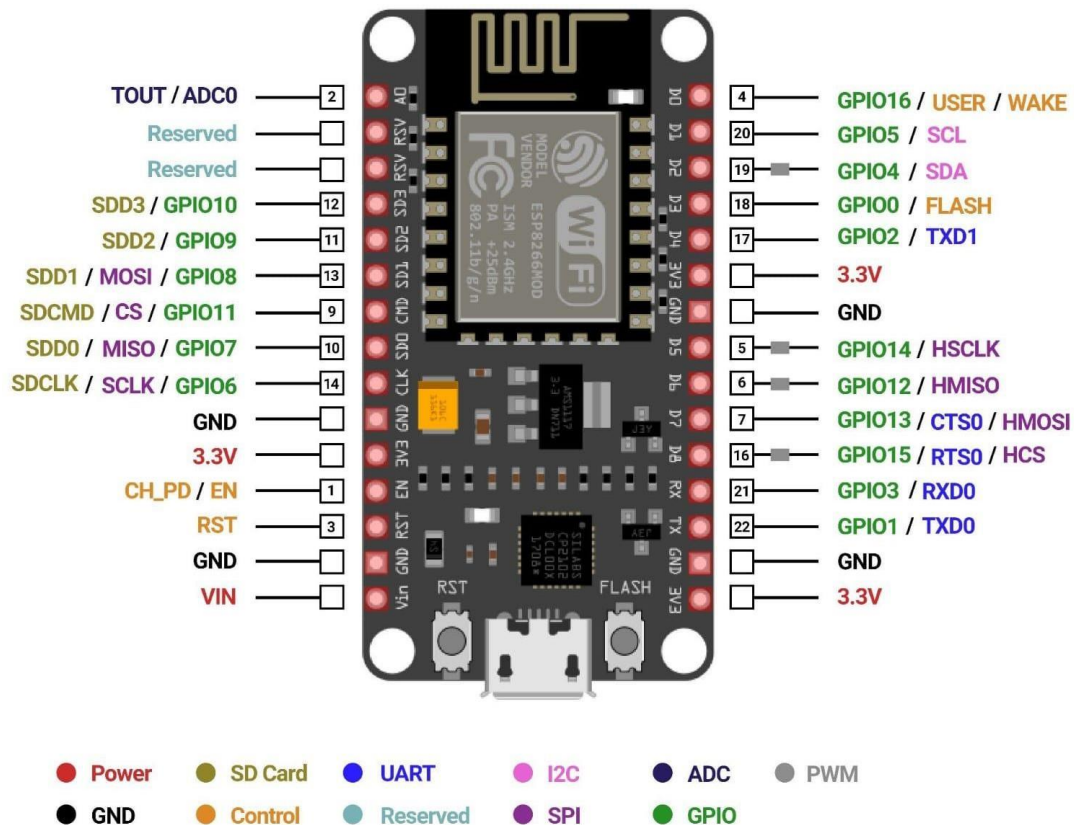


Figure3.2 NodeMCU ESP8266

NodeMCU ESP8266 Wifi Module is an open-source Lua based firmware and development board specially targeted for **IoT based applications**. It includes firmware that runs on the **ESP8266 Wi-Fi SoC** from **Espressif Systems**, and hardware which is based on the **ESP-12 module**.

Nodemcu ESP8266 Specifications & Features:

Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106

Operating Voltage: 3.3V

Input Voltage: 7-12V

Digital I/O Pins (DIO): 16

Analog Input Pins (ADC): 1

UARTs: 1

SPIs: 1

I2Cs: 1

Flash Memory: 4 MB

SRAM: 64 KB

Clock Speed: 80 MHz

USB-TTL based on CP2102 is included onboard, Enabling Plug n Play

PCB Antenna

Small Sized module to fit smartly inside your IoT projects

Nodemcu ESP8266 Pinout:

For practical purposes **ESP8266 NodeMCU** V2 and V3 boards present identical pinouts. While working on the NodeMCU based projects we are interested in the following pins.

Power pins (3.3 V).

Ground pins (GND).

Analog pins (A0).

Digital pins (D0 – D8, SD2, SD3, RX, and TX – GPIO XX)

Most ESP8266 NodeMCU boards have one input voltage pin (Vin), three power pins (3.3v), four ground pins (GND), one analog pin (A0), and several digital pins (GPIO XX).

Pin	Code	Arduino alias
A0	A0	A0
ssD0	GPIO 16	16
D1	GPIO 5	5
D2	GPIO 4	4
D3	GPIO 0	0
D4	GPIO 2	2
D5	GPIO 14	14
D6	GPIO 12	12
D7	GPIO 13	13
D8	GPIO 15	15
SD2	GPIO 9	9
SD3	GPIO 10	10
RX	GPIO 3	3
TX	GPIO 1	1

3.3 SENSORS USED IN THIS PROJECT

3.3.1 REES52 Soil Moisture Sensor

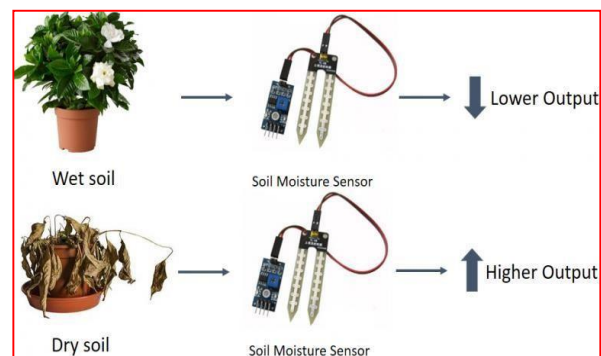
The soil moisture sensor is shown in the figure 3.5.(a) and the hygrometer is usually used to detect the humidity of the soil. So, it is perfect to build an automatic watering system or to monitor the soil moisture of the plants.

- The sensor is set up by two pieces: the electronic board (at the right), and the probe with two pads, that detects the water content (at the left).
- The sensor has a built-in potentiometer for sensitivity adjustment of the digital output (D0), a power LED and a digital output LED.

How does it work?



(a) Soil Sensor



(b) Function of Sensor

Figure 3.3 REES52 Soil Moisture Sensor.

The sensor is interfaced with Arduino microcontroller [7] and programmed. Once it is programmed it is placed inside a box and kept in the field. The soil moisture sensor has two probes which is inserted into the soil. The probes are used to pass current through the soil. The moisture soil has less resistance and hence passes more current through the soil whereas the dry soils has high resistance as shown in the figure 3.5.(b) and pass less current through the soil. The resistance value help detecting the soil moisture.

The voltage is that the sensor output changes accordingly to the water content in the soil and the connection diagram shown in the figure 3.6.

When the soil is Wet: the output voltage decreases. Dry: the output voltage increases.

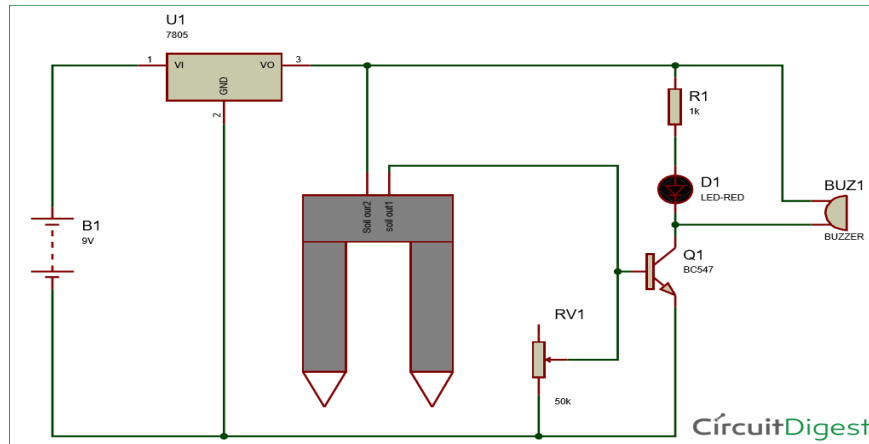


Figure 3.4 Connection diagram of Soil Moisture Sensor.

3.3.1 REES52 DHT11 Temperature and Moisture

Sensor Difference between DHT11 Sensor and module

The DHT11 sensor can either be purchased as a sensor or as a module. Either way, the performance of the sensor is same. The sensor will come as a 4-pin package out of which only three pins will be used whereas the module will come with three pins as shown above. The only difference between the sensor and module is that the module will have a filtering capacitor and pull-up resistor inbuilt, and for the sensor, you have to use them externally if required.

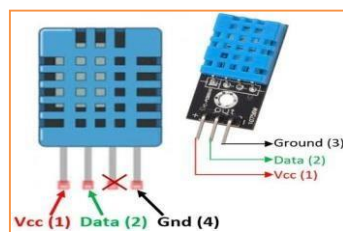
Where to use DHT11

The DHT11 is a commonly used Temperature and humidity sensor. The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data.

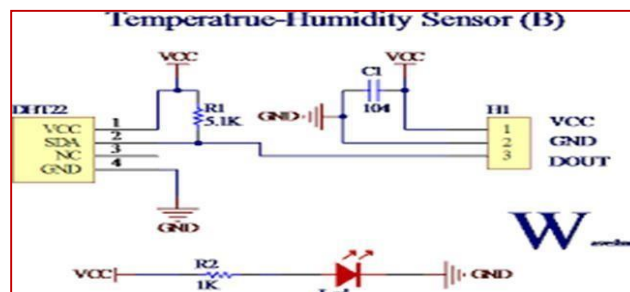
The sensor is also factory calibrated and hence easy to interface with other microcontrollers. The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of $\pm 1^\circ\text{C}$ and $\pm 1\%$. So if you are looking to measure in this range then this sensor might be the right choice for you.

How to use DHT11 Sensor

The DHT11 Sensor is shown in the figure 3.7.(a) and it is factory calibrated and outputs serial data and hence it is highly easy to set it up. The connection diagram for this sensor is shown below in the figure 3.7 (b).



(a) DHT11 Sensor



(b) Pin connection diagram

Figure 3.5 Temperature & Moisture Sensor.

DHT11 Specifications

- Operating Voltage: 3.5V to 5.5V
- Operating current: 0.3mA (measuring) 60uA (standby)
- Output: Serial data

- Temperature Range: 0°C to 50°C
- Humidity Range: 20% to 90%
- Accuracy: $\pm 1^\circ\text{C}$ and $\pm 1\%$
- Resolution: Temperature and Humidity both are 16 bit

DHT11 Pin out Details

Pin No	Pin Name	Description
1	Vcc	Power Supply 3.5V to 5.5V
2	Data	Output both Through serial data
3	NC	No Connection
4	Ground	Connected to Ground

3.8 PIR MOTION SENSOR

When the sensor is idle, both slots detect the same amount of IR, the ambient amount radiated from the room or walls or outdoors. When a warm body like a human or animal passes by, it first intercepts one half of the PIR sensor, which causes a positive differential change between the two halves. When the warm body leaves the sensing area, the reverse happens, whereby the sensor generates a negative differential change. These change pulses are what is detected .The PIR Sensor ,The IR sensor itself is housed in a hermetically sealed metal can to improve noise/ temperature/humidity immunity. There is a window made of IR- transmissive material (typically coated silicon since that is very easy to come by) that protects the sensing element. Behind the window are the two balanced sensors.



Figure 3.6 PIR SENSOR

3.9 PIRs Motion Sensors Working

PIR sensors are more complicated than many of the other sensors explained in these tutorials (like photocells, FSRs and tilt switches) because there are multiple variables that affect the sensors input and output. To begin explaining how a basic sensor works, we'll use this rather nice diagram. The PIR sensor itself has two slots in it, each slot is made of a special material that is sensitive to IR. The lens used here is not really doing much and so we see that the two slots can 'see' out past some distance (basically the sensitivity of the sensor).

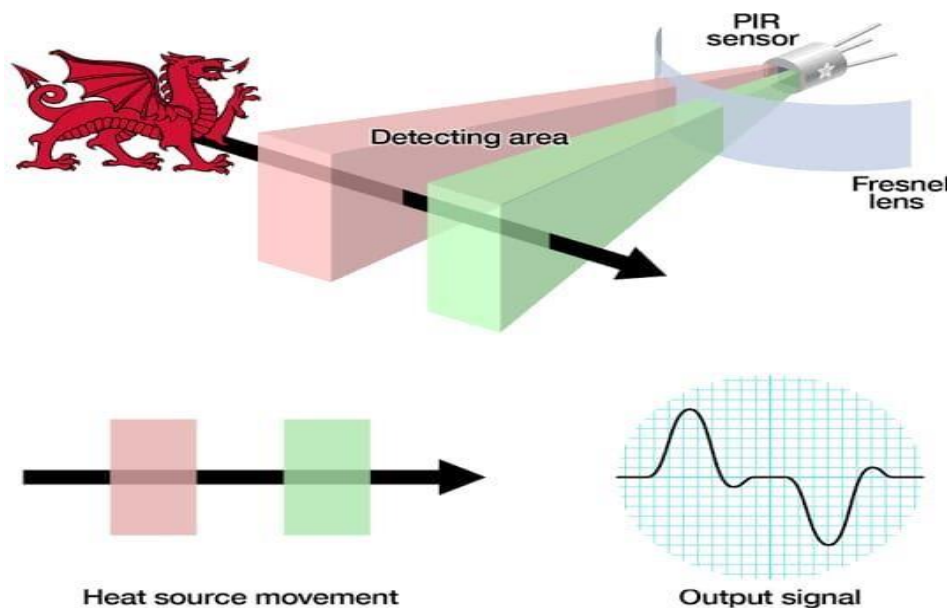


Figure 3.7 PIRs Motion Sensors

3.10 Rain drop sensor

Rain Sensor is a simple sensor that gives digital output. This sensor is very common for detecting rains, especially in tropical and subtropical regions.



Figure 3.8 Rain drop sensor

3.11 Rain drop sensor working

- The **working of the** rain sensor is very simple if you take a look at the schematic above. You'll see that it is a very simple circuit which just compares the voltages and according to that delivers the output.
- The Non-inverting input of one comparator is connected to the sensing plate with a 10 k Ω pull-up resistor. Also, it has 10nF capacitor in parallel to reduce the noise from the sensing plate. The other end of the plate is connected to GND.
- The inverting input of the same comparator with a 10k potentiometer between VCC & GND. This is done in order to control the value to the microcontroller. Hence, as a result, controlling the comparing the voltage of the two inputs of the comparator.
- The output pin of the comparator is connected to the OUT pin of the module, which gives value to the microcontroller. Also, it is connected to the on-board LED for indication in the **rain sensor**.

3.3 REES52-2 CHANNEL 5V RELAY MODULE AND MOTOR CONTROL

- Relays are most commonly used switching device in electronics. Let us learn how to use one in our circuits based on the requirement of our project.
- Before we proceed with the circuit to drive the relay we have to consider two important parameter of the relay.

One is the Trigger Voltage, this is the voltage required to turn on the relay that is to change the contact from Common-

- >NC to Common->NO. Our relay here has 5V trigger voltage, but you can also find relays of values 3V, 6V and even 12V so select one based on the available voltage in your project. The other parameter is your Load Voltage & Current, this is the amount of voltage or current that the NC, NO or Common terminal of the relay could withstand, in our case for DC it is maximum of 30V and 10A. Make sure the load you are using falls into this range.

REES52-2 CH 5V Relay module Pinout Details

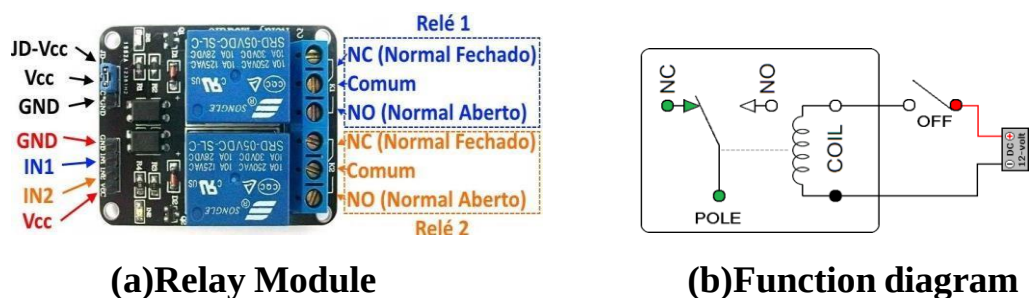


Figure 3.9 5V Relay Module Pinout Diagram

The above figure 3.12. (a) is the relay module pinout details and the figure 3.12.(b) circuit shows a bare-minimum concept for a relay to operate. Since the relay has 5V trigger voltage we have used a +5V DC supply to one end of the coil and the other end to ground through a switch. This switch

can be anything from a small transistor to a microcontroller or a microprocessor which can perform switching operating. You can also notice a diode connected across the coil of the relay, this diode is called the Fly back Diode. The purpose of the diode is to protect the switch from high voltage spike that can produced by the relay coil. As shown one end of the load can be connected to the Common pin and the other end is either connected to NO or NC. If connected to NO the load remains disconnected before trigger and if connected to NC the load remains connected before trigger.

Features of 5-Pin 5V Relay

- Trigger Voltage (Voltage across coil) : 5V DC
- Trigger Current (Nominal current) : 70mA
- Maximum AC load current: 10A @ 250/125V AC
- Maximum DC load current: 10A @ 30/28V DC
- Compact 5-pin configuration with plastic moulding
- Operating time: 10msec Release time: 5msec
- Maximum switching: 300 operating/minute (mechanically)

3.3.1 Applications of Relay for Motor control

- Commonly used in switching circuits.
- For Home Automation projects to switch AC loads
- To Control (On/Off) Heavy loads at a pre-determined time/condition is shown in figure 3.13.
- Used in safety circuits to disconnect the load from supply in event of failure
- Used in Automobiles electronics for controlling indicators glass motors etc.

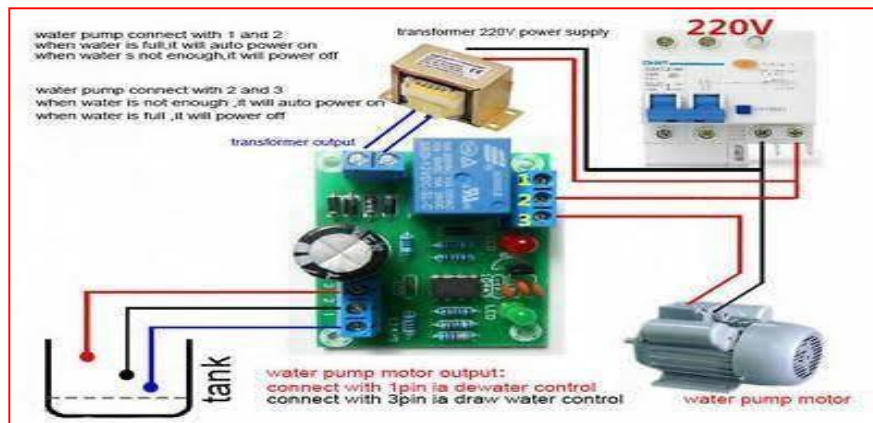


Figure 3.10 5V Relay and Motor control circuit.

3.4 APPLICATION OF IoT FOR BLYNK APP

Blynk is a hardware-diagnostic IoT platform [5] with white-label mobile apps, private clouds, device management, data analytics, and machine learning.

Figure 3.14 shows the IoT platform to connect your devices to the cloud, design apps to control them, analyze telemetry data, and manage your deployed products at scale

Blynk makes complex IoT technology simple:

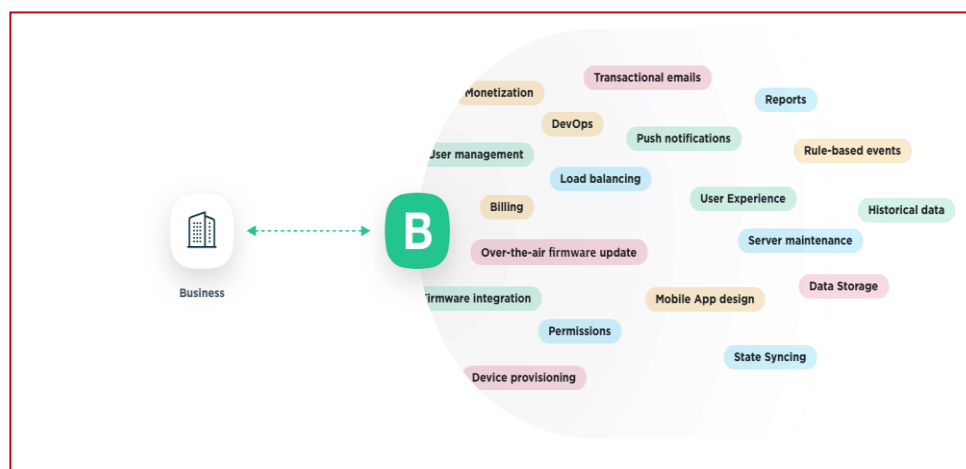


Figure 3.11 IoT Technology using Blynk App.

The most comprehensive IoT stack

- Connect any device to open and secure cloud.
- Choose from over 400 hardware models that work out-of-the-box with fast and open-source Blynk Cloud. Easily connect them over Wi-Fi, LTE, 2G-4G, or Ethernet

Drag-n-drop beautiful IoT apps in minutes

Choose from a variety of pre-designed widgets to build native iOS and Android apps that control electronics, monitor sensor data, and get notifications, and much more control your hardware from anywhere in the world which is shown in figure 3.15.



Figure 3.12 IoT Applications using Blynk App.

Turn your prototypes into commercial IoT solutions

In one click: the figure 3.16 which is shown below to export your project into a beautiful branded app that already works with your products. Then publish it to App Store[™] and Google Play[™]

Add your app icon, company logo, change colors, fonts, and publish to app stores as your own app. It's a white-label solution

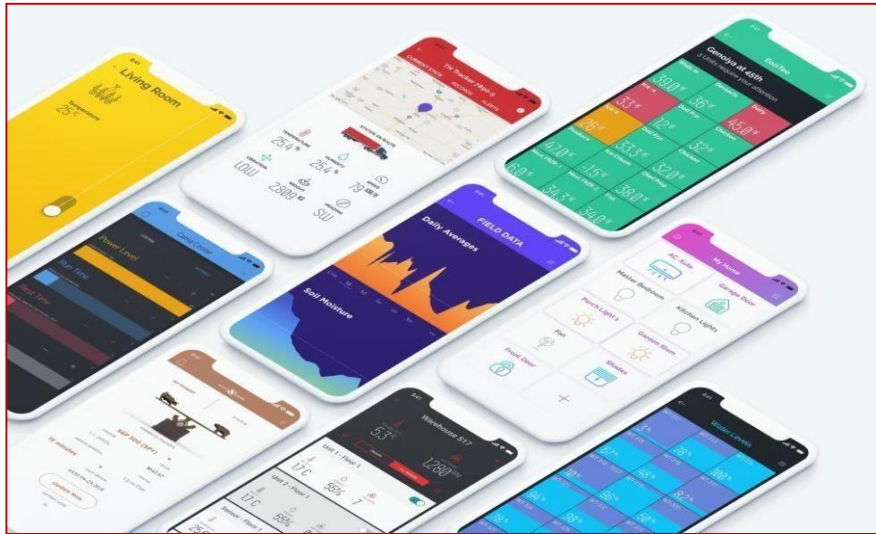


Figure 3.13 Different protocols using Blynk App.

Rgt4Manage your connected business in real-time

Get business insights, set access levels, manage end-users, devices, organizations, payments, ship firmware updates to your fleet over-the-air, and use machine learning algorithms is shown in figure 3.17.



Figure 3.14 Real time graphical using Blynk App.

Manage any number of devices, extend product lines, update firmware over-the-air, get notified about critical issues

A multi-tenancy environment with configurable access levels. Manage clients, distributors, contractors, installers, and anyone you work with and Getbusiness insights on multiple levels: from devices telemetry to a detailed stats and reports on your fleet of devices.

Automated data analysis for your business: forecasting, anomalies detection, pattern detection, pattern anomalies detection, and more.

3.5 SUMMARY

The chapter deals with the importance of IOT and its implementation using Blynk Application for the project of Smart Irrigation System and sensors used for monitoring the importance parameters and the ESP 8266 WeMos D1 - WIFI module used for communicating the data between end users.

CHAPTER 4

HARDWARE IMPLEMENTATION OF SMART IRRIGATION SYSTEM USING IoT

4.1FUNCTIONAL DIAGRAM OF HARDWARE

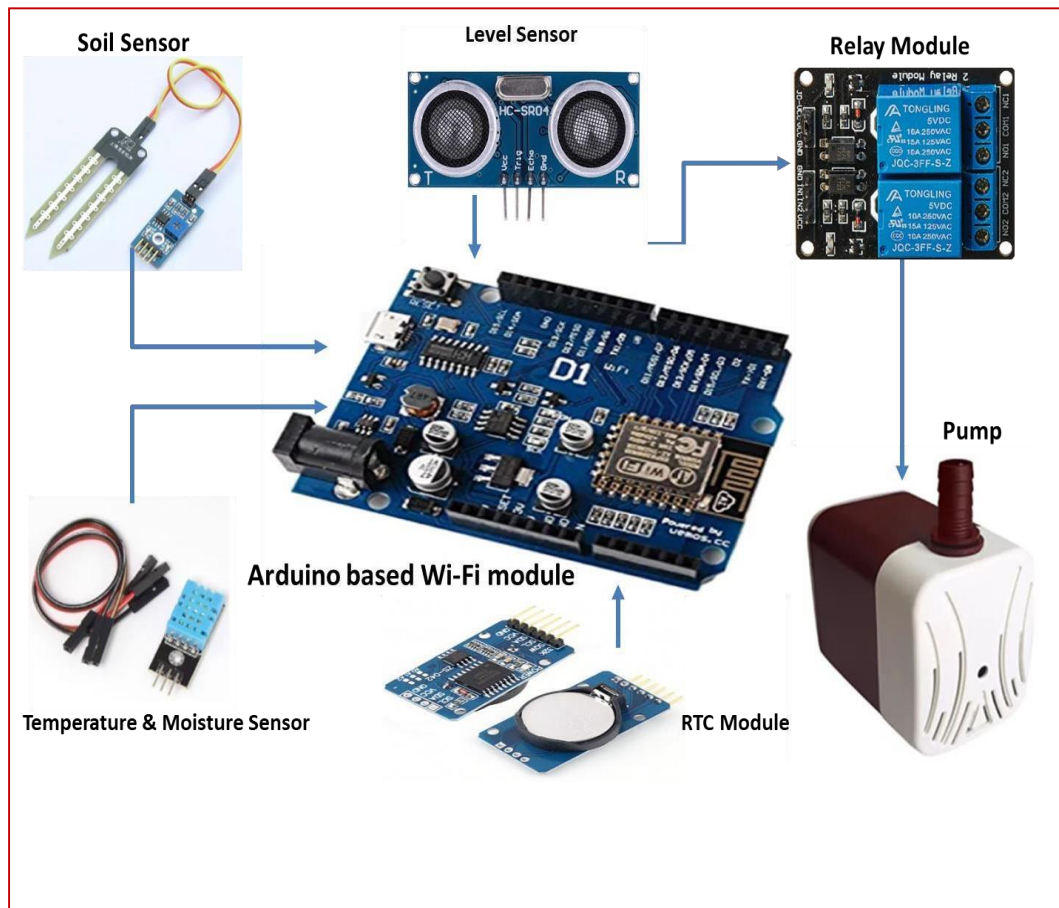


Figure 4.1 Functional diagram of Hardware used in SIS

The components used in this project are shown above in the figure 4.1 and the components, sensors and modules used in this project are described as below.

Soil Sensor

The soil moisture sensor [8] or the hygrometer is usually used to detect the humidity of the soil. So, it is perfect to build an automatic watering system or to monitor the soil moisture of the plants.

- The sensor is set up by two pieces: the electronic board (at the right), and the probe with two pads, that detects the water content (at the left).
- The sensor has a built-in potentiometer for sensitivity adjustment of the digital output (D0), a power LED and a digital output LED.

Circuit Diagram:

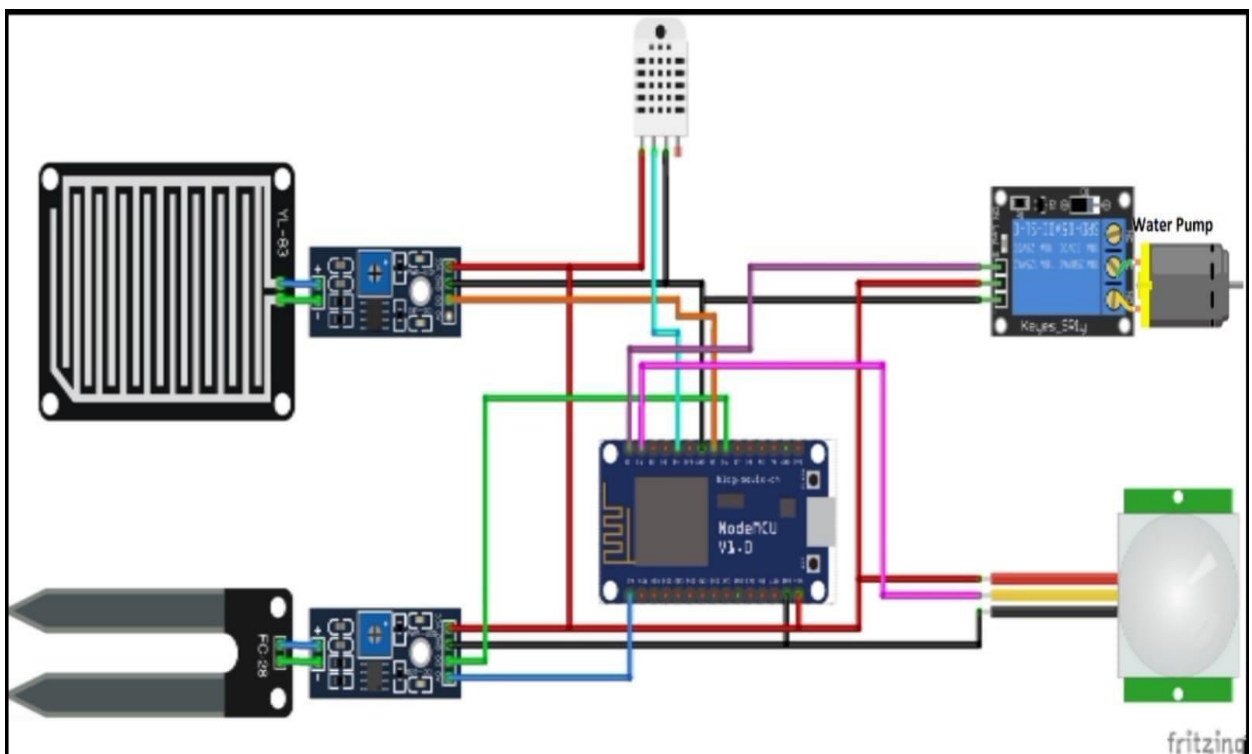


Figure 4.2 Circuit Diagram

Circuit Diagram explanation:

- Connect the VCC (power) and GND (ground) pins of the microcontroller to the respective power and ground lines on the breadboard.
- Connect the digital output pin of the soil moisture sensor to a digital input pin on the microcontroller (e.g., Pin 2).

- Connect the VCC and GND pins of the soil moisture sensor to the respective power and ground lines on the breadboard.
- Connect the VCC and GND pins of the water pump to the respective power and ground lines on the breadboard.
- Connect one terminal of the water pump to the NO (normally open) terminal of the relay module.
- Connect the COM (common) terminal of the relay module to the VCC line on the breadboard.
- Connect the GND pin of the relay module to the GND line on the breadboard.
- Connect the IN1 pin of the relay module to a digital output pin on the microcontroller (e.g., Pin 3).

DHT11 Temperature & Moisture Sensor

The DHT11 is a commonly used Temperature and humidity sensor [9]. The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of $\pm 1^\circ\text{C}$ and $\pm 1\%$. So if you are looking to measure in this range then this sensor might be the right choice for you.

5V Relay and Pump Circuit

The relay has 5V trigger voltage we have used a +5V DC supply to one end of the coil and the other end to ground through a switch. This switch can be anything from a small transistor to a microcontroller or a microprocessor which can perform switching operation. As shown one end of the load can be connected to the Common pin and the other end is either connected to NO or NC. If connected to NO the load remains disconnected before trigger and if connected to NC the load remains connected before trigger.

ALGORITHM

- STEP 1: Start the program.
- STEP 2: Insert the sensors in the soil.
- STEP 3: Initialize the Temperature sensor, Water level sensor and RTCmodule
- STEP 4: Initialize the Wi-Fi code to connect the Wi-Fi module.
- STEP 5: Make coding to Arduino to obtain the output based on Blynk input.
- STEP 6: Connecting the program with Blynk Application.
- STEP 7: When the outputs obtained are in the range/limit, and with input from Blynk is processed the motor output and parameters which need to be displayed.
- STEP 8: Then the output is uploaded to the Blynk App and required pump output is getting on by relay.
- STEP 9: Thus the output values are recorded in the form of graph in Blynk Application
- STEP 10: Repeat/Rescan the program by serial monitoring with Blynk cloud.

4.2 IMPLEMENTATION USING ARDUINO CODING:

```
// Viral Science www.viralsciencecreativity.com  
www.youtube.com/c/viralscience  
// Blynk IOT Smart Agriculture Monitoring System  
#define BLYNK_PRINT Serial  
#include <SPI.h>  
#include <ESP8266WiFi.h>
```

```

#include <BlynkSimpleEsp8266.h>
#include <SimpleTimer.h>
#include <DHT.h>
#define BLYNK_PRINT Serial
#include <OneWire.h>
#include <DallasTemperature.h>
#define ONE_WIRE_BUS D2
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);

char auth[] = "-----";           //Authentication code sent by Blynk
char ssid[] = "-----";           //WiFi SSID
char pass[] = "-----";           //WiFi Password

#define pirPin D1
int pirValue;
int pinValue;
#define sensorPin D6
#define rainPin D5
int sensorState = 0;
int rainState = 0;
int lastState = 0;
int lastRainState = 0;
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
SimpleTimer timer;

```

```

BLYNK_WRITE(V0)
{
  pinValue = param.asInt();
}

void sendSensor()
{
  float h = dht.readHumidity();
  float t = dht.readTemperature();

  if (isnan(h) || isnan(t)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }

  Blynk.virtualWrite(V5, h); //V5 is for Humidity
  Blynk.virtualWrite(V6, t); //V6 is for Temperature
}

void setup()
{
  Serial.begin(9600);
  Blynk.begin(auth, ssid, pass);
  pinMode(sensorPin, INPUT);
  pinMode(rainPin, INPUT);
  pinMode(pirPin, INPUT);
  dht.begin();
  timer.setInterval(1000L, sendSensor);
  Serial.begin(115200);
}

```

```

Blynk.begin(auth, ssid, pass);
  sensors.begin();
}
int sensor=0;
void sendTemps()
{
  sensor=analogRead(A0);
  sensors.requestTemperatures();
  float temp = sensors.getTempCByIndex(0);
  Serial.println(temp);
  Serial.println(sensor);
  Blynk.virtualWrite(V1, temp);
  Blynk.virtualWrite(V2,sensor);
  delay(1000);
}
void getPirValue(void)    //Get PIR Data
{
  pirValue = digitalRead(pirPin);
  if (pirValue)
  {
    Serial.println("Motion detected");
    Blynk.notify("Motion detected");
  }
}

void loop()
{

```

```

Blynk.run();
timer.run();
sendTemps();
sensorState = digitalRead(sensorPin);
Serial.println(sensorState);
if (sensorState == 1 && lastState == 0) {
  Serial.println("needs water, send notification");
  Blynk.notify("Water your plants");
  lastState = 1;
  delay(1000);
//send notification
}
else if (sensorState == 1 && lastState == 1) {
  //do nothing, has not been watered yet
  Serial.println("has not been watered yet");
  delay(1000);
}
else {
  //st
  Serial.println("does not need water");
  lastState = 0;
  delay(1000);
}
rainState = digitalRead(rainPin);
Serial.println(rainState);
if (rainState == 0 && lastRainState == 0) {
  Serial.println("Its Raining!");
  Blynk.notify("Its Raining!");
}

```



```

lastRainState = 1;
delay(1000);
//send notification
}
else if (rainState == 0 && lastRainState == 1) {
delay(1000);
}
else {
Serial.println("No Rains");
lastRainState = 0;
delay(1000);
}
if (pinValue == HIGH)
{
getPirValue();
}
Blynk.run();
delay(100);
}

```

4.3 HARDWARE IMPLEMENTATION:

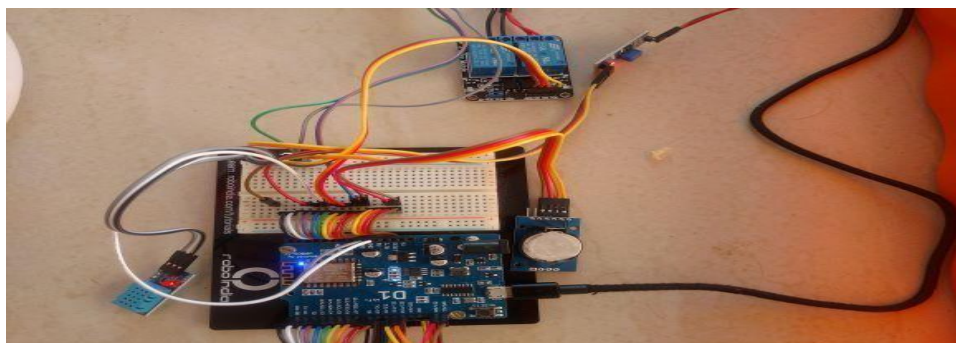


Figure 4.3 Hardware Implementation with all Sensor

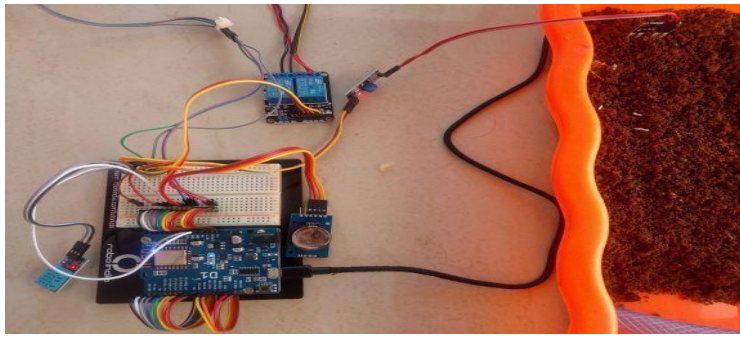


Figure 4.4 Hardware Implementation with soil sensor in sand

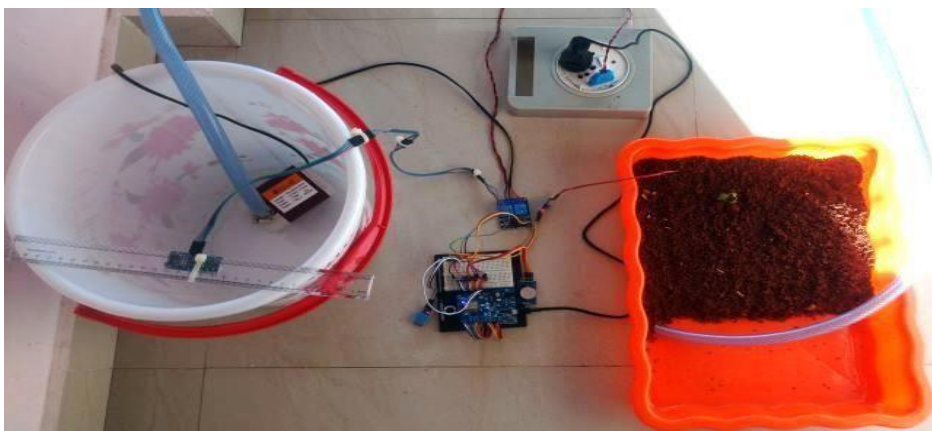


Figure 4.5 Hardware Implementation with complete assembly

The hardware implementation of the proposed Smart Irrigation System model is to switch on the pump for irrigation using smart phone and we can see the soil status in the mobile using Blynk Application. The output obtained is done by either manual mode or by Auto mode using the sensors with WeMos D1 ESP 8266 Arduino Module, Wi-Fi and IOT Technology.

▪ **Testing the Hardware kit**

- The ESP 8266 Module is connected with 5V Adapter using of mobile charger unit.
- Required Soil Sensor, Water level Sensor, Temperature & Humidity sensor boards and RTC module connected to the Arduino module as per its pin configuration as shown in the figure 4.2.

- Then through the Wi-Fi the Arduino Program is downloaded to the Arduino Module by the Arduino Software.
- The input from the Blynk App is first tested with LED output by both manual and Auto mode.
- The input of Soil sensor is tested with small amount soil when it is Dry and Wet condition as shown in the figure 4.3.
- The input of Temperature & Humidity sensor is verified with the Blynk App in the Smart Phone.
- RTC module function also verified in the Blynk App in the Smart Phone.
- Then Required Relay module is connected to Arduino board and the relay outputs connected to the testing single phase pump to the sample demo land unit which is shown in the figure 4.4.
- The Ultra sonic water level sensor is placed over the water tank as shown in the figure 4.4.
- Now the Pump is switched by the Smart Phone in both Manual mode and as well as in Auto mode.
- The pump is getting ON while pressing the manual ON button and also in Auto mode the pump is getting ON and OFF by the soil sensor output when it is Dry.

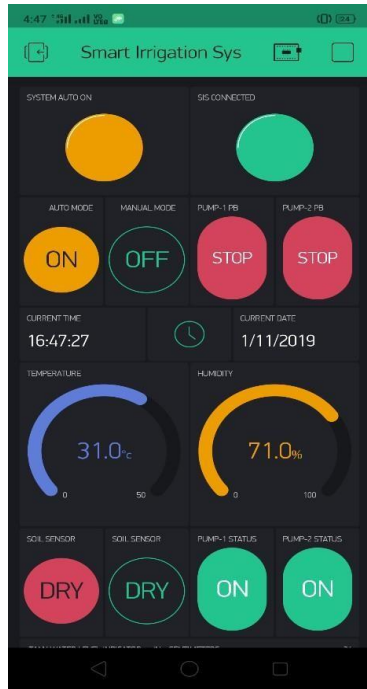
4.4 BLYNK OUTPUT RESPONSE AND GRAPH



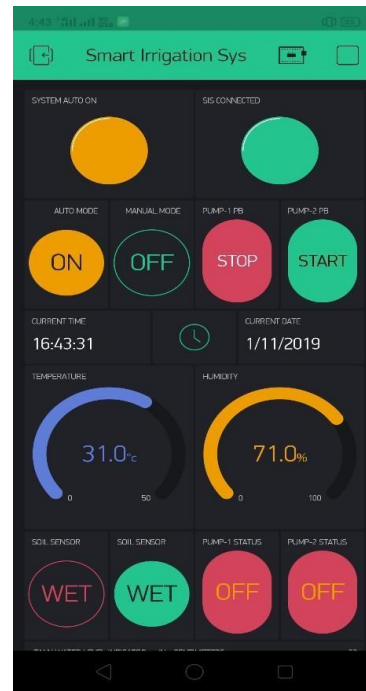
(a) Manual mode



(b) Manual mode On status



(c) Auto Mode On Staus



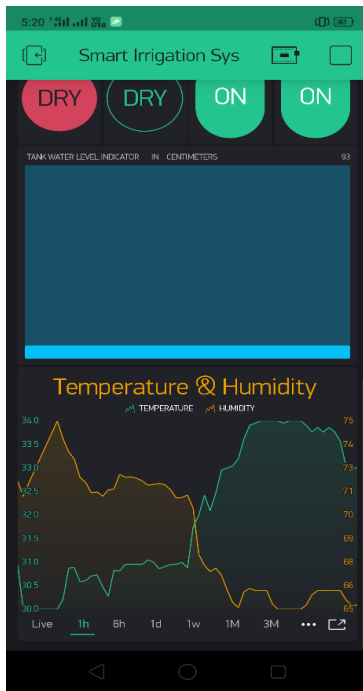
(d) Auto Mode Soil status

Figure 4.6 Manual and Auto Output Response in Blynk App

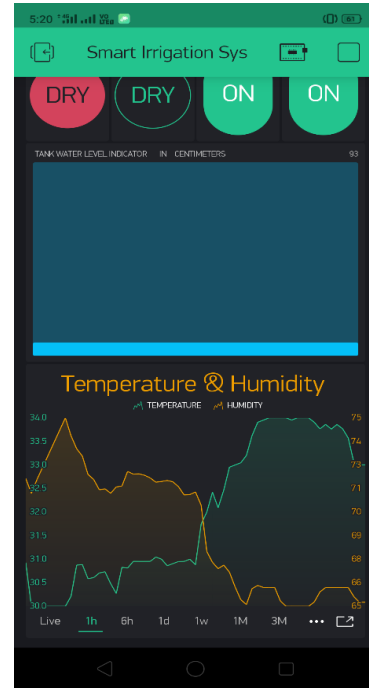
Smart Phone Input and Output status

- The Blynk Application is opened in the Smart phone and through the Hotspot it is connected to Arduino board.
- Then from this Blynk App the pump 1&2 ON input is pressed in Manual Mode and seen the status of the output -1&2 status in the Smart phone which is shown in the figure 4.5.(a) & 4.5.(b).
- Then the Auto mode is selected and seen the status of output 1&2 as shown in the figure 4.5.(c).
- The status of the Soil sensor input also can be seen in the Smart phone in the figure 4.5.(d).
- The actual Temperature and humidity can be measured in the form of digital value as well as graphical value as shown in the figure 4.6.(a).

- The tank water level [10] indicator is shown in the picture when it is in Maximum before Switch ON the Pump and Minimum level after pump switched ON in Auto mode is shown in the figure 4.6.(a) & (b).



(a) Water level full status



(b) Water level empty status

Figure 4.7 Water Level and Temperature & Humidity Graph

CHAPTER 5

CONCLUSION

Smart irrigation is an important technology in saving water for the plant which requires more attention especially in Iraq. Using WSN as an Internet of things gives more flexibility in system monitoring and management. Results show that the proposed network can be used along the season with minimum maintenance.

The proposed Smart Irrigation System for agriculture is that the farmer can operate the motor by remotely in his smart phone anywhere in the world either in manual mode or in Auto mode and he can verify the soil status and also the temperature of the motor by the graph in the mobile phone itself.

The proposed irrigation system for agricultural purpose can measure the Soilmoisture, temperature of the field and transmits the real time data to the user through the Wi-Fi and IoT server, if there is any deviation from the span of reference value, then the user can send the command through the IoT server to maintain the set point value of field parameter for a proper irrigation and proposed IoT based irrigation system is better than the recently proposed other irrigation systems.

6. REFERENCES

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